



Functional Requirement Document

EMERGENCY DESCENT function

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1 Executive summary

This document is the functional specification of the EMERGENCY DESCENT function. It provides “aircraft level” requirements that will be the main input for systems specifications.

The purpose of the EMERGENCY DESCENT function is to have the aircraft descend to a safe altitude in the case of loss of cabin pressurisation.

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2 Record of revisions

Issue	DATE	Reason for revision
V01	09/03/2026	First issue

3 General

3.1 Objective

This document is the functional specification for the EMERGENCY DESCENT function.

- Section 4 provides a high-level description of the EMERGENCY DESCENT function.
- Section 5 describes operational scenarios of the EMERGENCY DESCENT function.
- Section 6 contains "aircraft level" requirements for the EMERGENCY DESCENT function.

3.2 Applicability

[1] is the baseline for this functional specification. Thus this document applies to any aircraft described by [1].

3.3 Reference documents

[1] Baseline avionics description.

[2] Arinc 702a-5: Advanced flight management computer system. 2018.

3.4 Acronyms

3.5 Definitions

AP	Autopilot
ATC	Air Traffic Control
ATHR	Auto Thrust
V _{MO}	Minimum Operating Speed
ND	Navigation Display
FCU	Flight Control Unit
IAS	Indicated Air Speed
MSA	Minimum Sector Altitude
TBD	To Be Determined
SPD	Speed
HDG	Heading
TERR	Terrain

4 Function Description

The purpose of the EMERGENCY DESCENT function is to navigate the aircraft from its current altitude to 10 000 ft or the MSA, following the fastest safe descent profile.

The EMERGENCY DESCENT function provides the following features:

- It allows the flight crew to activate the EMERGENCY DESCENT.
- Upon activation of the EMERGENCY DESCENT:

The aircraft begins descending to 10 000 ft, or the published MSA (whichever is greater) at 250 kts IAS.

It allows the flight crew to select a desired altitude if different the altitude selected by the EMERGENCY DESCENT function.

It allows the flight crew to set a desired IAS for the descent.

- Upon arriving to the selected altitude, the aircraft levels off in ALT mode: the EMERGENCY DESCENT function is finished.
- The EMERGENCY DESCENT function can be deactivated at any time by selecting another vertical navigation AP function.

Note: The flight crew is responsible for extending the airbrakes to increase the vertical speed of the descent. They are also responsible for the retraction of the airbrakes when the aircraft levels off.

5 Operational Scenarios

In nominal scenario 1, it is an Emergency descent with structural damage, it is assumed that both pilots remain conscious throughout the emergency.

In nominal scenario 2, it is an Emergency descent without structural damage, it is assumed that both pilots remain conscious throughout the emergency.

5.1 Nominal scenario 1

Initial conditions: The aircraft is flying at a typical A320 cruise altitude, both pilots are in the cockpit, the autopilot and auto-thrust are engaged. An explosive depressurization occurs resulting in structural damage to the aircraft.

STEP 1 : A visual and an audio alert inform the pilots of the situation.

STEP 2 : The pilots put on their oxygen mask.

STEP 3 : The system sets :

1. the altitude to 10 000 ft
2. the speed to 250 kts
3. the heading to Selected Mode
4. the SPD/MACH knob is set to SPD

STEP 4 : There is an audio announcement of these actions, and the information is displayed on the FCU. The actions will be active (Managed Mode) after a delay and Terrain mode will be displayed on the ND. Meanwhile, pilots can still modify the instructions before they become active.

STEP 5 : the system alerts the air traffic controller of the situation (Emergency depressurization and Emergency descent) and of the target altitude and speed.

STEP 6 : pilots manage the systems and communications with ATC. They deploy

5.2 Nominal scenario 2

Initial conditions: The aircraft is flying at a typical A320 cruise altitude, both pilots are in the cockpit, the autopilot and auto-thrust are engaged. An explosive depressurization

occurs without structural damage to the aircraft.

STEP 1 : A visual and audio alert informs the pilots of the situation.

STEP 2 : The pilots put on their oxygen mask.

STEP 3 : The system sets :

1. the altitude to 10 000 ft
2. the speed to 250 kts
3. the heading to Selected Mode
4. the SPD/MACH knob is set to SPD

STEP 4 : There is an audio announcement of these actions, and the information is displayed on the FCU. The actions will be active (Managed Mode) after a delay and Terrain mode will be displayed on the ND. Meanwhile, pilots can still modify the instructions before they become active.

STEP 5 : pilots set speed to selected mode and increase it to $VMO + \Delta$.

STEP 6 : the system alerts the air traffic controller of the situation (Emergency depressurization and Emergency descent) and of the target altitude and speed .

STEP 7 : pilots manage the systems and communications with ATC. They deploy the speed brakes as needed.

5.3 Additional scenarios

None

(ideas for other scenarios : during climb, descent, spurious activation, ...)

6 Requirements

EMER DES - 001

The system shall enable the aircraft to reach an altitude of 10000 ft or MSA, whichever is higher, at maximum vertical speed, without any intervention of the pilot, in less than TBD minutes.

Rationale:	The system shall operate in any terrain environment, whether the pilots are operational or not.
Comments:	

EMER DES - 002

The system shall maintain the IAS of the aircraft at 250 kts throughout the whole descent if no pilot override has been provided.

Rationale:	In case of structural damage, the maximum speed would be 250 kts. Otherwise, the pilots, if conscious, could manually increase it to VMO - delta.
Comments:	

EMER DES - 003

The system shall set the HDG to Selected Mode and SPD/MACH to SPD in less than TBD seconds after the initial system activation.

Rationale:	As stated in the EMER DES procedures.
Comments:	

EMER DES - 004

The system shall display TERR on ND after the delay in less than TBD seconds.

Rationale:	Terrain should be displayed for situational awareness. This display should happen after the delay, when descent is initiated to avoid overwhelming pilots with information.
Comments:	Whether the system should act accordingly to the terrain will be discussed at a later stage.

EMER DES - 005

The pilots shall be informed visually and audibly of each action that the system is intending to perform.

Rationale:	Consequence of the pilot interview. The pilots should be aware of each and every action performed automatically by the system. In case of smoke in the cockpit (explosive decompression), the information should be audible as well.
Comments:	

EMER DES - 006

The pilots shall be able to cancel any spurious activation of the system with a single action.

Rationale:	The system should be easily cancellable if spurious.
Comments:	

EMER DES - 007

The pilots shall be able to modify any parameter set by the system with a single action.

Rationale:	The system should be easily modifiable by pilots (change of HDG, speed increased to VMO-delta,...).
Comments:	

EMER DES - 008

The system shall inform the ATC of the EMER DES mode of the aircraft, the target parameters as well as the current vertical speed of the aircraft.

Rationale:	
Comments:	We will see at a later stage if we could use ADS-C or CPDLC for specifying the requirements for ATC communications.

EMER DES - 009

The system shall be disengaged as soon as the aircraft is stabilised at 10000 ft or MSA, whichever is higher.

Rationale:	The system should be disabled when stabilisation is reached, at the lowest “breathable” altitude.
Comments:	

EMER DES - 010	
The system shall be disengaged as soon as any other vertical navigation mode is enabled.	
Rationale:	If the pilots decide to override the system, or when another vertical mode becomes active, the EMER DES system can be disabled.
Comments:	

7 Open items

8 Appendices